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(D) Oily water system

Table of contents

(D) OILY WATER SYSTEM **1**
1 SCOPE AND RELATED DOCUMENTS **2**
2 ABBREVIATIONS **2**
3 OILY WATER SYSTEM **2**
3.1 OILY WATER COLLECTION 3
3.2 OILY WATER SUMPS NO. 3-5 4
3.3 ALCO SUMPS NO. 1-7 4
3.4 WORKSHOP OILY WATER SUMP 4
3.5 OILY WATER TRANSFER PUMP UNITS DAD 903-904 & 907-912 & 914-915 5
3.5.1 Function 5
3.5.2 Control 5
3.6 OILY WATER TRANSFER PUMP UNIT (MANUAL) DDA 901 5
3.6.1 Function 5
3.6.2 Control 6
3.7 OILY WATER BUFFER TANK DAB 901 6
3.8 OILY WATER TREATMENT UNIT DBB 901 AND OW FEED PUMP 6
3.9 SLUDGE TANK DDB 901 7
3.10 SLUDGE TRANSFER PUMP UNIT DDD 901 7

1 Scope and related documents

This document describes the basic control and functionality of the oily water treatment system. It is to be read in conjunction with the latest revision of main flow diagrams listed in Table 1.

The description covers primarily control functions and operations which are not integrated on the engine. Typical WFI tank instrumentations and controls have been described in order to get acquainted with the working entity of the oily water treatment system.

Table 1. Reference documents

Document number	Title, main flow diagram
DBAF489727	(D) Oily water treatment system 1(3), main flow diagram
DBAF489726	(D) Oily water treatment system 2(3), main flow diagram
DECA0003909	(D) Oily water treatment system 3(3), main flow diagram
DETA00005122	MOD0_1 EAM 3.0 1-C ALCO
DETA00005127	PCC 0_1 Engine fuel booster unit 3.0 ALCO
DETA00005124	MOD90_ ALCO pipe rack 3.0
DBAF509447	DAD 903-904 & 907 OW transfer pump unit Automatic, main flow diagram
DBAE159406	DAD 908-912 & 914-915 OW transfer pump unit Automatic, main flow diagram
DBAF509461	DDA 901 OW transfer pump unit Manual, main flow diagram
DAAF537524	DBB 901 OW treatment unit, main flow diagram
DAAF398067	DAD 913 OW treatment feed pump unit, main flow diagram
DBAF509461	DDD 901 Sludge transfer pump unit Manual, main flow diagram

2 Abbreviations

OW Oily Water

PLC Programmable Logic Controller

OWT Oily Water Treatment

3 Oily water system

The oily water system comprises of OW collection, storage, pumping and treatment. OW is collected into the OW sumps and pumped to OW buffer tank DAB 901 or directly to sludge tank DDB 901. Oily water from OW buffer tank is pumped via OWT feed pump to oily water treatment unit DBB 901. Sludge from the sludge tank is pumped into tanker trucks for disposal via sludge transfer unit DDD 901. With few exceptions all piping is insulated.

Main equipment in oily water treatment system:

Equipment	Code	Scope
Oily water transfer pump units	DAD 901-904 & 907	WFI
Oily water transfer pump units (for ALCO)	DAD 908-912 & 914-915	WFI
Oily water buffer tank	DAB 901	WFI
Oily water treatment unit (incl. feed pump)	DBB 901	WFI
Oily water transfer pump (manual)	DDA 901	WFI
Boiler washing water tank	DAD 905	WFI
Boiler washing water pump	DAD 906	WFI
Sludge tank	DDB 901	WFI
Sludge transfer pump unit	DDD 901	WFI

3.1 Oily water collection

Oily water is collected in the sumps from the following locations/consumers in the power plant area:

- drain from engine hall
- drain from fuel treatment house
- drain from workshop
- drain from unloading station
- drain from transfer area
- drain and overflow from day tank and storage tank area
- Oily water is also collected from separate sludge tanks in the following units:
    - Engine fuel booster units:
        - Lube oil separator units
        - Dirty leak tank
- ALCO separator units

The drainage from the different sources flows by gravity to the oily water sumps around the site. The sumps are divided into Oily water sumps and ALCO sumps. The purpose of this split is to keep the drains that are associated with ALCO (Ex classified) separated from drains and equipment that are not.

The day tank and storage area have separate drainage channels for rain water. Clean rainwater is lead to rain water drainage. The valves on the piping for rain water drain are to be kept closed and only opened under supervision.

All drain valves from the tank area are to be kept closed and only opened under supervision. This is due to the large potential flow of fluids in case of emergency. If the valves were open the OW sumps would quickly be overfilled.

3.2 Oily water sumps No. 3-5

The oily water sumps are underground cast concrete sumps.

OW sumps are equipped with multi-level level switches for controlling oily water transfer pump units DAD 903-904 and 907 with the following functionality when activated:

LS^{HH} High-high level alarm, Alarm in WOIS

LS^{H} High level alarm, OW pump start

LS_{L} Low level alarm, OW pump stop

The pump is automatically started when the high level is achieved.

Automatic pump stop occurs for:

- DAD 903 when low level is reached in sump No. 3
- DAD 904 when low level is reached in sump No. 4
- DAD 907 when low level is reached in sump No. 5

The sumps are also equipped with a strainer for pump protection from debris and other bigger solids.

3.3 ALCO sumps No. 1-7

The ALCO sumps are underground cast concrete sumps. The sumps are used in locations where ALCO is expected to be part of the drain collection, and is therefore collected in its own dedicated ALCO sump.

ALCO sumps are equipped with multi-level level switches for controlling oily water transfer pump units DAD 908-909-910-911-912 and 914-915 with the following functionality when activated:

LS^{HH} High-high level alarm, Alarm in WOIS

LS^{H} High level alarm, OW pump start

LS_{L} Low level alarm, OW pump stop

The pump is automatically started when the high level is achieved.

Automatic pump stop occurs for:

- DAD 910 when low level is reached in sump No. 1
- DAD 909 when low level is reached in sump No. 2
- DAD 911 when low level is reached in sump No. 3
- DAD 912 when low level is reached in sump No. 4
- DAD 908 when low level is reached in sump No. 5
- DAD 914 when low level is reached in sump No. 6
- DAD 915 when low level is reached in sump No. 7

The sumps are also equipped with a strainer for pump protection from debris and other bigger solids.

3.4 Workshop oily water sump

The workshop oily water sump has the same construction as the OW sumps. However, the instrumentation consists only of a high-high level switch connected to WOIS. The sump is emptied

directly to disposal truck or to sludge tank manually by workshop oily water transfer pump unit DDA 901.

The water drained from the workshop often contains surfactants, which can emulsify oil making the separation process in the oily water treatment unit difficult and inefficient.

3.5 Oily water transfer pump units DAD 903-904 & 907-912 & 914-915

3.5.1 Function

The OW transfer pumps are diaphragm pumps for pumping the oily water from the sumps (either from OW sumps or ALCO sumps) to the OW buffer tank DAB 901 or to the sludge tank DDB 901.

The OW transfer pump units consist mainly of the following components:

Component	Function
Pump, D001	OW from sump to buffer tank or sludge tank
Solenoid valve, V002	Operating air for pump

The OW transfer pump units also have shut out valves on inlet and outlet in addition to a non-return valve. A pressure indicator is installed on the outlet of the pump. The functional principle is identical for emptying of the OW sumps and ALCO sumps.

3.5.2 Control

The oily water transfer pump units are equipped with control panels. The control is normally set to “auto” mode, which means that the pumps start and stop automatically according to the low and high level signals from the level switch in the corresponding oily water sump.

Oily water pumps are used to empty sumps. When the high level switch is reached in sump the pump is started and will be run until low level switch is activated in sump.

The pumps can also be used manually in case a sump needs to be emptied. This operation needs to be supervised by an operator.

The collected OW is pumped to the OW buffer tank. In case the oily water sump is loaded with high amounts of oil, it is possible to pump it directly to the sludge tank. The choice is made by manually turning the three-way valve (remote control), DAD 900-V001, towards the sludge tank. This operation is to be done under controlled conditions.

3.6 Oily water transfer pump unit (manual) DDA 901

3.6.1 Function

The OW transfer pump is a diaphragm pumps for pumping the oily water from the sump to a disposal truck or to sludge tank.

The OW transfer pump units consist mainly of the following components:

Component	Function
Pump, D001	OW from sump to truck
Manual valve, V002	Operating air for pump

The OW transfer pump units also have shut out valves on inlet and outlet in addition to a non-return valve and a pressure indicator on the outlet side.

3.6.2 Control

The workshop OW sump is equipped with a level switch (DDA 900 L001, LSHH) that only sends a high high level alarm to the WOIS. When the high high level alarm is received the operator must start the OW transfer pump, DDA 901 manually in order to empty the sump.

The contents of the sump are emptied into a truck. The reason for keeping this OW separate from the rest of the system is that the OW from the workshop contains surfactants. The surfactants can emulsify oil making the separation process in the oily water treatment unit difficult and inefficient.

3.7 Oily water buffer tank DAB 901

The tank serves as an intermediate storage for OW and also allows for some homogenisation.

The temperature and the level of the OW buffer tank can be read locally and on WOIS.

The OW buffer tank is equipped with three level switches with the following suggested functionality when activated:

LS^{HH}	High-high level alarm, Alarm in WOIS
LS^{H}	High level alarm, Alarm in WOIS
LS_{L}	Low level alarm, stop signal to OW feed pump
LT	Level transmitter, remote level monitoring

The tank has an emergency overflow and a drain.

Water can and should be drained from the OW buffer tank manually in order to remove free water. The drain water is led back to oily water sump No. 2.

3.8 Oily water treatment unit DBB 901 and OW feed pump

The oily water treatment unit consist of pumps, two flotation tanks, a chemical dosing system, filters and a sludge collecting tank. The sludge from the unit is pumped to the sludge tank (DDB 901) by a built in sludge pump while the treated water is disposed to an open pit from where it is disposed.

The OW feed pump is equipped with a viscosity sensor to measure the oil content of the influent. If oil content is to high the flow is diverted to the sludge tank.

A sensor will also analyse the effluent water, and direct it back to an OW sump for re-entry to the water treatment system in case the required water quality is not achieved.

For further information about the control and function of the OWT unit please refer to the supplier’s manual.

3.9 Sludge tank DDB 901

The temperature and the level of the sludge in the sludge tank can be read locally from the thermometer and level indicator. A temperature transmitter is also connected to WOIS.

The sludge tank is equipped with three level switches with the following functionality when activated:

LS^{HH} High-high level alarm: alarm in WOIS, stop OW feed pump

LS^{H} High level alarm: alarm in WOIS, manual emptying to be started (DDD 901 pump)

LS_{L} Low level alarm: alarm in WOIS, manual emptying to be stopped (DDD 901 pump)

The tank has an emergency overflow and a drain.

The high-high level alarm warrants immediate attention from the operators as the OW treatment unit will be stopped by the alarm. The OW buffer tank is not emptied when the OWT unit is stopped. To avoid any risk of tank overflows or sump overflows, the sludge tank should always be emptied in time.

Water can and should be drained from the sludge tank manually before disposal in order to remove free water. The drain water is led to the valve pit.

3.10 Sludge transfer pump unit DDD 901

When the sludge tank DDB 901 is to be emptied to the truck, the manual sludge transfer pump unit DDD 901 is used. The hose is connected to the container truck with quick couplings and the pump is started manually by opening the compressed air supply valve. Also, the manual valve (DDD 900 V002) located after the sludge transfer pump unit must be opened.

The pump is stopped when the container truck is full, by closing the air supply valve. The valve (DDD 900 V002) must be manually closed.

An operator should always be standby when the sludge tank is emptied.